

Species-Aware Module Review: Fatty Acid β -Oxidation Spiral in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI 160488; proteome UP000000556)

Module reviewed: Mitochondrial fatty acid β -oxidation spiral (cross-species generic module)

Local pathway bucket: KEGG `ppu00071` "Fatty acid degradation" **Prepared for:** manual module satisfiability and gene-annotation curation

1. Executive summary

- The generic module is a **eukaryotic/mitochondrial** construct. In the bacterium *P. putida* KT2440 the identical four-step chemistry is executed in the **cytoplasm** by the **FadBA fatty-acid-oxidation multienzyme complex** plus standalone acyl-CoA dehydrogenases — **not** by a mitochondrial trifunctional protein (MTP). The module is **satisfiable by conserved enzyme families**, but should be curated as *covered-by-bacterial-analog*, not mapped subunit-for-subunit onto HADHA/HADHB/ECHS1/HADH/ACAA2.
- **All four obligate steps + the upstream activation feeder are covered.** No step is a genuine gap.
- **Evidence tier is thin.** Of the 22 candidates, **only fadB (PP_2136) and fadA (PP_2137) are Swiss-Prot "reviewed"**; the other 20 are unreviewed TrEMBL, and **all 22 have UniProt protein-existence = "Inferred from homology."** There is no protein-level experimental annotation for any candidate; functional support comes from KT2440 physiology/engineering studies, not per-gene biochemistry.
- **$\approx$ Half of the candidate list is over-propagation** from neighboring CoA-thioester pathways (glutaryl-CoA, sulfinopropanoyl-CoA, phenylacetyl-CoA, alcohol dehydrogenases, alkane-hydroxylase redox partners, and EC 2.3.1.9 PHA/acetoacetyl thiolases) and should **not** count toward spiral coverage.
- **The candidate metadata has omissions.** A proteome-wide EC scan (UniProt, taxon 160488) shows the true standalone step-3 3-hydroxyacyl-CoA dehydrogenase (EC 1.1.1.35) is **PP_2214**

(Q88KS5), absent from the 22-gene list, and a third fatty-acyl-CoA ligase PP_2038 (Q88L97) is likewise missing. PP_2214–PP_2217 form a **contiguous four-gene β -oxidation cassette** (steps 3-4-1-2), so several genes previously read as "scattered over-annotation" are actually a coherent paralogous operon.

2. Target-organism pathway definition

Included process (working scope): the iterative, cytoplasmic degradation of saturated acyl-CoA esters two carbons per turn, comprising four obligate reactions — (1) FAD-dependent α,β -dehydrogenation (acyl-CoA dehydrogenase, EC 1.3.8.x), (2) hydration of *trans*-2-enoyl-CoA (enoyl-CoA hydratase, EC 4.2.1.17), (3) NAD⁺-dependent oxidation to 3-ketoacyl-CoA (3-hydroxyacyl-CoA dehydrogenase, EC 1.1.1.35), (4) thiolytic cleavage to acetyl-CoA + chain-shortened acyl-CoA (3-ketoacyl-CoA thiolase, EC 2.3.1.16). The chain-shortened acyl-CoA re-enters step 1 (the spiral). An upstream **activation feeder** (long-chain-fatty-acid-CoA ligase, EC 6.2.1.3) is required to make the acyl-CoA substrate.

Keep separate (neighboring pathways / overview maps): - De novo **fatty-acid biosynthesis** (fab genes; ACP-linked, EC 2.3.1.- synthases) — opposite direction; a documented source of confusion in *P. putida* (cryptic FabF2,).

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- **mcl-PHA biosynthesis** (phaC/phaG/phaJ) — draws (R)-3-hydroxyacyl-CoA *from* the β -oxidation spiral but is a distinct module ().

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- **Unsaturated-FA auxiliary cassette** (Δ^3,Δ^2 -enoyl-CoA isomerase; $\Delta^3,5,\Delta^2,4$ -dienoyl-CoA isomerase; 2,4-dienoyl-CoA reductase) — explicitly out of scope for the *saturated* spiral. - **Alkane oxidation (alk system)** — upstream of β -oxidation; supplies fatty acids but is not part of the spiral (relevant to mis-bucketed alkT/PP_5371). - Amino-acid / aromatic / alicyclic acyl-CoA catabolism (glutaryl-CoA, phenylacetyl-CoA, cyclohexanecarbonyl-CoA) — separate KEGG buckets.

Alternate names / DB definitions: KEGG [ppu00071](#) "Fatty acid degradation"; EC-level it overlaps E. coli **fad** regulon nomenclature (fadD, fadE, fadBA, fadL, fadR). Bacterial FadB $\equiv$ eukaryotic "trifunctional enzyme α " by homology only ().

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3. Expected step model (as realized in KT2440)

Step	Chemistry (EC)	Eukaryotic players (module)	KT2440 realization
Activation (feeder)	acyl-CoA synthetase (6.2.1.3)	—	fadD-I/II PP_4549/PP_4550 (+ dispersed fadD paralogs)
1. α,β -dehydrogenation	acyl-CoA dehydrogenase (1.3.8.7/1.3.8.8)	VLCAD/MCAD/SCAD	fadE PP_1893 (+ PP_2437, PP_2793, PP_3725)
2. hydration	enoyl-CoA hydratase (4.2.1.17)	HADHA (MTP- α), ECHS1	fadB PP_2136 (bifunctional α); ancillary PP_2217
3. NAD ⁺ oxidation	3-hydroxyacyl-CoA DH (1.1.1.35)	HADHA (MTP- α), HADH	fadB PP_2136 (same polypeptide) + standalone PP_2214
4. thiolysis	3-ketoacyl-CoA thiolase (2.3.1.16)	HADHB (MTP- β), ACAA2, ACAT1	fadA PP_2137 (+ paralog PP_2051)

Key architectural point: steps 2 + 3 are fused in one multifunctional FadB polypeptide (EC 4.2.1.17 + 1.1.1.35 + 5.1.2.3 epimerase + 5.3.3.8 isomerase), and FadB (PP_2136) + FadA (PP_2137) are adjacent — a **fadBA operon** homologous to the *E. coli* $\alpha_2\beta_2$ complex (

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Second architectural point (proteome scan): a distinct **contiguous cassette PP_2214–PP_2217** supplies each activity as *separate* polypeptides — PP_2214 (3-hydroxyacyl-CoA dehydrogenase type-2, EC 1.1.1.35 → step 3), PP_2215 (thiolase EC 2.3.1.9 → step 4-type), PP_2216 (*acd*, acyl-CoA dehydrogenase family → step 1-type), PP_2217 (enoyl-CoA hydratase EC 4.2.1.17 → step 2). Its *acd* (cyclohexanecarbonyl-CoA / sulfinopropanoyl-CoA) annotation suggests specialization for alicyclic/branched or sulfur-containing acyl-CoAs, but it is a real β -oxidation-type operon, not noise. **PP_2214 is the only standalone EC 1.1.1.35 step-3 gene besides fadB and is missing from the candidate list.**

Full proteome EC complement (taxon 160488): step 1 = 3 ACADs (fadE PP_1893, PP_2437, PP_2793); step 2 = fadB, PP_2217, paaF (paa); step 3 = fadB, **PP_2214**, paaH PP_3282 (paa-specific); step 4 = fadA PP_2137, fadA PP_2051, bktB; activation = fadD PP_4549, PP_4550, **PP_2038**.

4. Candidate genes and evidence

Core spiral — retain (high confidence): - **fadB** PP_2136 / Q88L02 — *Swiss-Prot reviewed*. Fatty-acid-oxidation complex subunit α ; covers **steps 2 & 3** (+ auxiliary epimerase/isomerase for unsaturated substrates). Highest-confidence assignment. → promote. - **fadA** PP_2137 / Q88L01 — *Swiss-Prot reviewed*. 3-ketoacyl-CoA thiolase (β subunit); **step 4**, EC 2.3.1.16. Operonic with fadB. → promote. - **fadE** PP_1893 / Q88LN6 — acyl-CoA dehydrogenase EC 1.3.8.7/1.3.8.8; primary **step 1** candidate; primary bucket already ppu00071. Homology-only. → promote. - **fadD-I** PP_4549 / **fadD-II** PP_4550 — EC 6.2.1.3; linked **activation** pair analogous to *P. aeruginosa* fadD1/fadD2 (long- vs short-chain;

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→ promote both.

Core spiral — retain (paralogs, medium confidence): - **PP_2437 / PP_2793** — long-chain acyl-CoA dehydrogenases (EC 1.3.8.8); plausible chain-length isozymes for **step 1**. - **fadA** PP_2051 / Q88L84 — thiolase I (EC 2.3.1.16); plausible second **step-4** thiolase. - **PP_2217 / Q88KS2** — enoyl-CoA hydratase (EC 4.2.1.17); possible standalone **step-2** ancillary (crotonase-family). - **PP_3725 / Q88GJ7** — "acyl-CoA dehydrogenase," *no EC assigned*; weak step-1 candidate, needs review.

Physiological context (direct KT2440 evidence that the spiral operates): fatty acids and alkane-derived octanoate/decanoate are catabolized through β -oxidation, and its intermediate *trans*-2-enoyl-CoA feeds mcl-PHA synthesis (

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This supports module activity even though per-gene biochemistry is absent.

5. Gaps, ambiguities, and likely over-annotations

Not the core saturated spiral — reassign / candidate_uncertain: - **gcdH** PP_0158 (glutaryl-CoA dehydrogenase, EC 1.3.8.6) — Lys/Trp degradation (ppu00380). - **acd** PP_2216 (3-sulfinopropanoyl-CoA desulfonase / cyclohexanecarbonyl-CoA dehydrogenase, EC 1.3.8.11/3.13.1.4) — sulfur/alicyclic CoA catabolism (ppu00410). - **paaF** PP_3284 (enoyl-CoA

hydratase-isomerase) — phenylacetate (paa) catabolism. - **frmA PP_1616** (S-hydroxymethylglutathione/formaldehyde dehydrogenase, EC 1.1.1.1/1.1.1.284) and **adhP PP_3839** (short-chain alcohol dehydrogenase, EC 1.1.1.1) — alcohol dehydrogenases, **not** the EC 1.1.1.35 3-hydroxyacyl-CoA dehydrogenase of step 3. Classic over-propagation via the shared "1.1.1.-" root. - **alkT PP_5314** (rubredoxin-NAD⁺ reductase, EC 1.18.1.1) and **PP_5371** (rubredoxin/rubredoxin reductase) — electron-transfer partners of the **alkane hydroxylase** system, upstream of β -oxidation, not spiral enzymes. - **EC 2.3.1.9 thiolases** — PP_2215, PP_3355, PP_0582, **yqeF PP_4636**, **bktB PP_3754**: acetoacetyl-CoA / PHA-biosynthetic thiolases (condensing direction). The degradative spiral thiolase is **EC 2.3.1.16**; only fadA/PP_2051 qualify. bktB in particular is a known PHA-pathway β -ketothiolase.

Missing from candidate metadata (should be ADDED): - **PP_2214 (Q88KS5)** — 3-hydroxyacyl-CoA dehydrogenase type-2, EC 1.1.1.35: the genuine standalone **step-3** gene (candidate list had only the multifunctional fadB for this step; frmA/adhP were wrong). High priority to add. - **PP_2038 (Q88L97)** — a third long-chain-fatty-acid-CoA ligase (EC 6.2.1.3): additional **activation** feeder. - Note the **PP_2214–PP_2217 cassette** is coherent; PP_2215/PP_2216/PP_2217 are better curated as a specialized β -oxidation-type operon (candidate_uncertain, likely alicyclic/branched substrate) than as isolated over-annotations. - **paaH PP_3282** (EC 1.1.1.35) sits beside paaF PP_3284 → confirms a phenylacetate (paa) cluster; keep both out of the core spiral.

Ambiguities / caveats for curation: - Broad EC/GO on acyl-CoA-dehydrogenase and thiolase superfamilies drives false membership; filter step-1 by EC 1.3.8.7/1.3.8.8 and step-4 by EC 2.3.1.16. - "Reviewed" status of fadBA is UniRule-based, **not experimental for KT2440**; treat as high-confidence homology, not proof. - Chain-length specialization of fadE/PP_2437/PP_2793 and fadD-I/II is inferred from *Pseudomonas* orthologs, not measured in KT2440.

6. Module and GO-curation recommendations

Module element	Recommended status	Rationale
Step 1 acyl-CoA dehydrogenase	covered	fadE PP_1893 + paralogs; multiple candidates, homology-strong
Step 2 enoyl-CoA hydratase	covered	fadB PP_2136 (Swiss-Prot), +PP_2217
Step 3 3-hydroxyacyl-CoA DH	covered	fadB PP_2136 (Swiss-Prot) + standalone PP_2214 (EC 1.1.1.35, add to metadata)
Step 4 3-ketoacyl-CoA thiolase	covered	fadA PP_2137 (Swiss-Prot) + PP_2051
Activation feeder (EC 6.2.1.3)	covered	fadD-I/II PP_4549/4550 + dispersed paralogs
Mis-bucketed dehydrogenases/thiolases/redox	candidate_uncertain → reassign	belong to gcdH/acd/paa/alk/PHA modules
Eukaryotic subunit mapping (MTP α/β)	module_needs_revision	bacterial FadBA $\neq$ MTP; add bacterial-analog note

- **Module boundary revision:** the generic module's subunit-level scaffold (HADHA/HADHB split, ECHS1/HADH vs MTP) is **wrong for bacteria**. Add a species-aware note that steps 2+3 are fused in one FadB polypeptide and step 4 is the operonic FadA; do not require MTP-specific GO terms.
- **GO terms:** existing GO MF terms suffice (GO:0016508/0016509-type acyl-CoA dehydrogenase, GO:0004300 enoyl-CoA hydratase, GO:0003857 3-hydroxyacyl-CoA dehydrogenase, GO:0003988 acetyl-CoA C-acyltransferase, GO:0004467 long-chain fatty-acid-CoA ligase). No new GO request appears necessary; use CC "cytoplasm," not mitochondrion.
- **No new module document needed** for the saturated spiral beyond the bacterial-analog annotation; the unsaturated auxiliary cassette remains separately curated.

7. Genes to promote to full fetch-gene review

1. **PP_2136 (fadB)** — anchor, steps 2+3; confirm operon and multifunctionality.
2. **PP_2137 (fadA)** — anchor, step 4.
3. **PP_1893 (fadE)** — primary step-1 dehydrogenase.
4. **PP_4549 / PP_4550 (fadD-I/II)** — activation pair; resolve chain-length roles vs *P. aeruginosa* fadD1/fadD2.
5. **PP_2051 (thiolase I)** and **PP_2437 / PP_2793** (long-chain acyl-CoA DHs) — resolve paralog redundancy and chain-length coverage.
6. **PP_2214 (Q88KS5)** — add to metadata and review as the standalone step-3 dehydrogenase; review the **PP_2214–PP_2217 cassette** as a unit to determine substrate specialization. Also add/review **PP_2038 (Q88L97)** as a third acyl-CoA ligase. (*Down-rank — reassign to their primary buckets rather than promote:*) gcdH, acd, paaF, frmA, adhP, alkT, PP_5371, PP_2215, PP_3355, PP_0582, yqeF, bktB.

8. Key references

- DiRusso CC (1990) *J Bacteriol* — E. coli fadBA operon; $\alpha_2\beta_2$ multienzyme complex; homology to eukaryotic trifunctional enzyme.

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- Kang Y et al. (2010) — multiple FadD acyl-CoA synthetases; fadD1 long-chain / fadD2 short-chain in *P. aeruginosa*.

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- Zarzycki-Siek J et al. (2013) — six FadD homologs in *P. aeruginosa*; chain-length division of labor.

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- Liu Q et al. (2023) *Appl Microbiol Biotechnol* — β -oxidation–PHA relationship in *P. putida* KT2440 revisited; PhaJ homologs.

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- Palacios-Ferrer A et al. (2026) — alkane→octanoate/decanoate enter β -oxidation in *P. putida*.

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- Sathesh-Prabu C et al. (2025) — *E. coli* vs *P. putida* KT2440 as fatty-acid cell factories.

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- Dong H et al. (2021) — cryptic FabF2 / FA-synthesis vs β -oxidation competition in *P. putida* F1.

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Uncertainty & species-transfer notes

- **Direct KT2440 evidence:** functional β -oxidation operates (physiology/engineering;

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Per-gene biochemistry is **absent**; all 22 annotations are homology-inferred.

- **Same-genus transfer (moderate–strong):** FadD chain-length specialization from *P. aeruginosa*

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- **Family-level transfer only (weak on subunit identity):** eukaryotic MTP/ACAD/thiolase subunit assignments — use EC/family, not orthology.